

SIJIE YANG

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Urban Analytics Lab, National University of Singapore, Singapore

EDUCATION

National University of Singapore, Singapore <i>Full-funded PhD Candidate. Research Direction: Smart City and Urban Informatics. Supervised by Assist Prof Filip Biljecki</i>	2023 - present
University of Pennsylvania (UPenn), Philadelphia, the United States <i>Full-funded Master of Computer Science and Information Technology; GPA 3.93 out of 4.00. Core Course: 5920 Discrete Mathematics; CIT 5930 CIT 5950 Computer System Programming; CIT 5940 Data Structure and Software Development; CIT 5960 Algorithms and Computation; CIS 5210 Artificial Intelligence; CIS 5810 Computer Vision & Computational Photography; CIS 5450 Big Data Analytics; ESE 5420 Stat for Data Science.</i>	2021 - 2025
University College London (UCL), London, the United Kingdom <i>Master of Science, Space Syntax: Architecture and Cities; Distinction, the Highest-Level Degree. Distinction Master Thesis. Course taken: BARC0015 Knowledge-Based Design; BARC0016 Buildings, Organizations and Networks; BARC0018 Spatial Cultures; BARC0019 Spatial Justice; BARC0020 Architectural Phenomena; BARC0024 Space Syntax Methodology and Analytical Design.</i>	2021 - 2022
Chongqing University (CQU), Chongqing, China <i>Bachelor of Architecture (5-year program); Top 5%; GPA 3.61 out of 4.00. Awarded Engineer of Excellence by Ministry of Education in China.</i>	2016 - 2021
University of California, Berkeley (UCB), California, the United States <i>Undergraduate Certificate. Awarded Excellent International Student.</i>	2018 - 2019

PROFESSIONAL EXPERIENCE

Urban Analytics Lab, National University of Singapore PhD Researcher, Supervised by Filip Biljecki <i>Doctoral research on the conceptualisation and quantification of urban spatial cognition, using multimodal AI methods (CV, VLMs, GNNs, neuromorphic models, RAG, contextual engineering) and diverse data sources (SVI, GIS, surveys, social media, mobility).</i>	2023 - present
NUS Cities, National University of Singapore Teaching Assistant in AR5323 Green and Well Built Environment, Supervised by Prof Lam Khee Poh <i>Taught modules on green and healthy building analysis, thermal comfort assessment and modelling, and built environment data analytics; assisted with assignments, exams, and student support.</i>	2023 - 2025
School of Engineering and Applied Science, University of Pennsylvania Teaching Assistant in Discrete Mathematics, Computer Architecture, Artificial Intelligence <i>Assisted teaching in Discrete Mathematics (graph theory, logic, combinatorics, probability), Computer Architecture (binary logic, CMOS, CPU design, assembly, C programming, hardware-software interface), and Artificial Intelligence (search, reasoning, planning, machine learning, NLP; Russell & Norvig, AIMA). Supported grading, online Q&A, and student consultations.</i>	2022 - 2025
Data Morphosis Lab, Cornell University, University of Hong Kong Research Assistant, Supervised by Assis Prof. Waishan Qiu, <i>Conducted urban morphology and GIS analysis; explored GNNs, vision-language models, and spatial regression for urban perception studies; supported research design, data collection, and paper writing.</i>	2022 - present
Key Laboratory of New Technology of Cities in Mountain Area, Chongqing University, Chongqing, China Research Assistant in Urban Soundscape Analysis, Supervised by Prof. Hui Xie, <i>Performed field surveys on urban soundscapes and acoustic materials; cleaned and modelled data using statistical methods; contributed to reports and manuscripts.</i>	Aug 2019 - Aug 2020
Institute of Architectural History and Theory, Chongqing University, Chongqing, China Research Assistant in Architectural Surveying, Supervised by Dr. Jiang Jialong, <i>Supported field surveys of historical architecture (Longwang Temple, Zigong); applied UAV imaging, point cloud scanning, and manual measurement; built 3D models and drafted architectural drawings.</i>	Jun 2019 - Jul 2019
Lacime Architects Co., LTD, Shanghai, China Architectural and Urban Design Intern, Supervised by Architect. Yin Tan, <i>Assisted design projects with sketches, 3D modelling, rendering, and construction drawings; prepared presentations and supported design logic development.</i>	Jul 2020 - Mar 2021

PUBLICATIONS

Yang, S. , Trivic, Z., Zhu, Y., Abdelrahman, M., & Biljecki, F. (2025). Urban comfort optimisation via social media and quantum simulation. <i>Nature Cities (Under Review)</i> .
Peng, C., Yang, S. (co-first) , Liu, A., Xiang, Y., Zhou, Z., & Biljecki, F. (2025). Mapping Subjective Window View Perception from Real Estate Images. <i>Landscape and Urban Planning (Under Review)</i> .
Yang, S. , Chong, A., Liu, P., & Biljecki, F. (2025). Seeing Heat for Climate Action: Feature Fusion of Street View Imagery in Thermal Affordance Mapping. <i>NeurIPS 2025 Workshop: Tackling Climate Change with Machine Learning (Under Review)</i> .
Yang, S. , Lei, B., & Biljecki, F. (2025). Urban Comfort Assessment in the Era of Digital Planning: A Multidimensional, Data-driven, and

AI-assisted Framework. In *19th International Conference on Computational Urban Planning and Urban Management (CUPUM 2025)*. UCL East, London, UK

Yang, S., Chong, A., Liu, P., & Biljecki, F. (2025). Thermal comfort in sight: Thermal affordance and its visual assessment for sustainable streetscape design. *Building and Environment*, 112569.

Yang, S., Krenz, K., Qiu, W., & Li, W. (2023). The role of subjective perceptions and objective measurements of the urban environment in explaining house prices in greater London: A multi-scale urban morphology analysis. *ISPRS International Journal of Geo-Information*, 12(6), 249.

Xu, X., Qiu, W., Li, W., Huang, D., Li, X., & **Yang, S.** (2022). Comparing satellite image and GIS data classified local climate zones to assess urban heat island: A case study of Guangzhou. *Frontiers in Environmental Science*, 10, 1029445.

Yang, S., & Yang, Y. (2022, August). Cultural impacts on traditional Chinese garden design: A configurational comparison between traditional Chinese imperial and private gardens using space syntax. In *Proceedings of The 13th International Space Syntax Symposium* (p. 414). International Space Syntax Symposium.

Yang, S., & Psarra, S. (2022). The social impact of atrium space in multilevel buildings: An analysis of university libraries using space syntax. Available at SSRN 4249348 (Preprint).

AWARDS & EXHIBITION

Tan Cheng Siong Research Scholarship , Singapore	Apr 2025
Granted by NUS, Singapore.	
NUS Research Scholarship , Singapore	Mar 2023
Granted by Ministry of Education, Singapore.	
MSc Graduated with Distinction , London, the United Kingdom	Dec 2022
Granted by University College London.	
B.Arch. Graduated with Engineer of Excellence in Chinese University Engineer Training Programme , Chongqing, China	Jun 2021
Granted by the Ministry of Education, China.	
The 6th China Human Environment Design Academic Year Award, Honourable Mention , Guangzhou, China	Oct 2020
Design competition for all the university students; Judged by The Architectural Society of China and Tsinghua University.	
<i>Urban design project with Wang Jie and Ren Yi, named "Inter Streetwise", as team leader and main designer.</i>	
The 5th CUC CUP Architecture Design International Competition, 3rd Place , Hangzhou, China	Oct 2019
Design competition for all the university students; Judged by The Architectural Society of China.	
<i>Architecture design project with Gao Yanan, museum design named "Back to Yi Nationality", as team leader and main designer.</i>	
Young Designers Award, Honourable Mention , Nanjing, China	Aug 2019
Design competition for China's undergraduate students; Granted by Southeast University.	
<i>Architecture design project with Gao Yanan, museum design named "Back to Yi Nationality", as team leader and main designer.</i>	
Architecture Biennale 2018, 16th International Architecture Exhibition, Art Document Exhibition of Chinese Teachers & Students in Architecture Majors of Colleges and Universities , Venice, Italy	Oct 2018
Recognized by National Steering Committee for Architecture (China);	
<i>Individual Architecture Painting "Summer" depicting Modern Art Museum, Shanghai, in 2018 summer.</i>	
National Olympiad in Informatics in Provinces (NOIP), First Prize , China	2013
Granted by China Computer Federation (CCF).	